

THE SOUL OF MHNKAN

Evolution, Philosophy, and the EML Paradigm

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The Stacking Trap

We build massive, towering networks simply because we do not know the exact mathematical complexity required to compose them.

Towering Stacks of Linear Projections



The Functional Escape

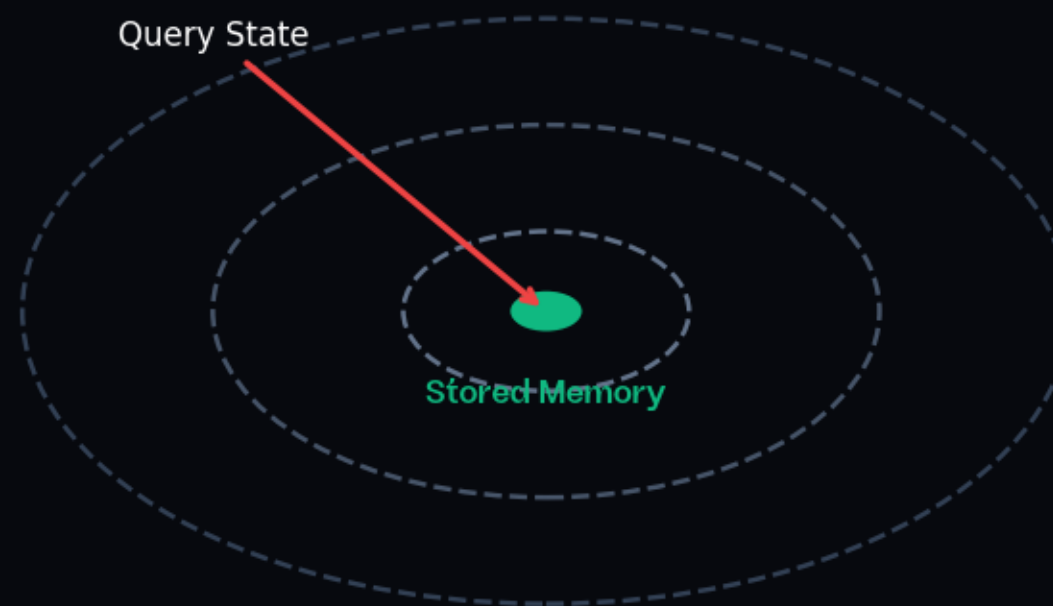
Kolmogorov–Arnold Networks map continuous pathways directly, replacing parameter weight stacks with smooth mathematical flow.

Continuous Function Pathways



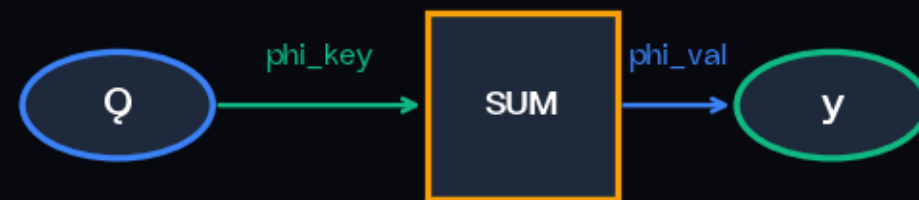
Storing Memory as Physics

Continuous associative memory behaves like a gravity well, where queries slide naturally into stable attractor basins.



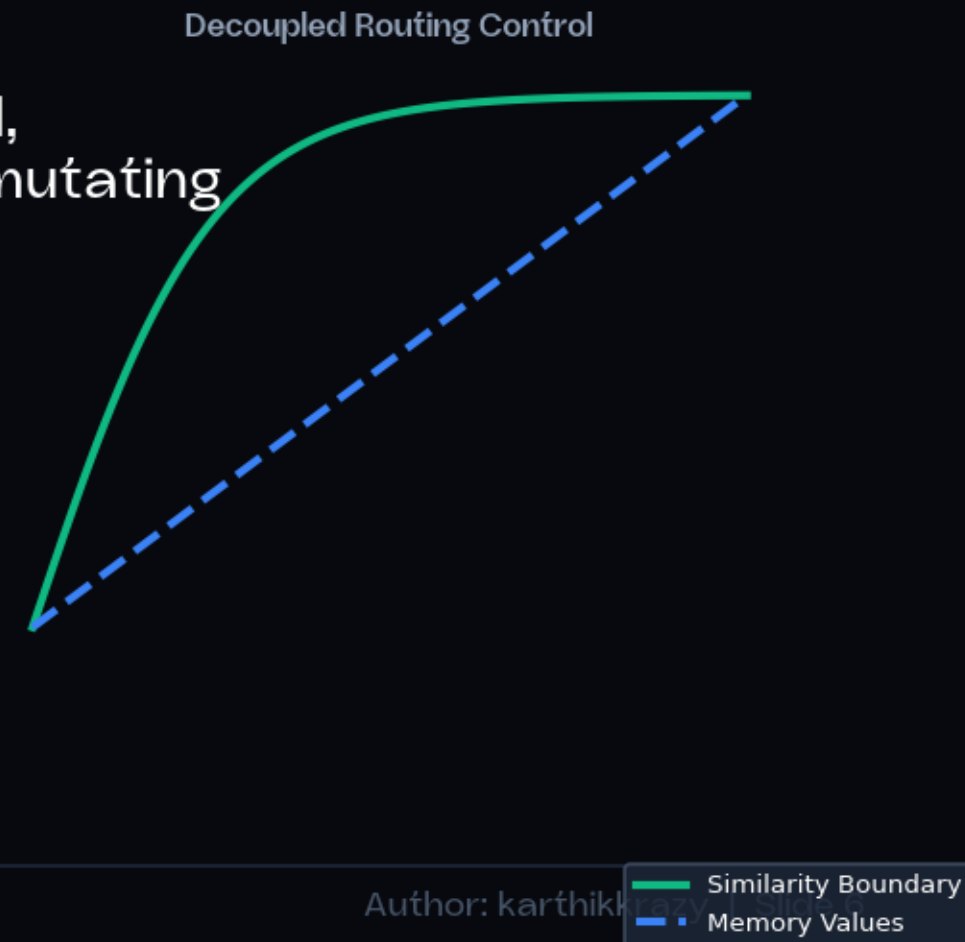
The MHN - KAN Mapping

We project memory queries onto a two-layer Kolmogorov-Arnold Network to route data dynamically through sum nodes.



Decoupling Similarity

By separating pattern matching from retrieval, we can adjust attraction boundaries without mutating the stored memory values.



Zero - Training Memorization

Analytical Projection Matrix

We didn't train this network—we just directly computed the weights using Hopfield rules. This is the nature of Modern Hopfield memory.



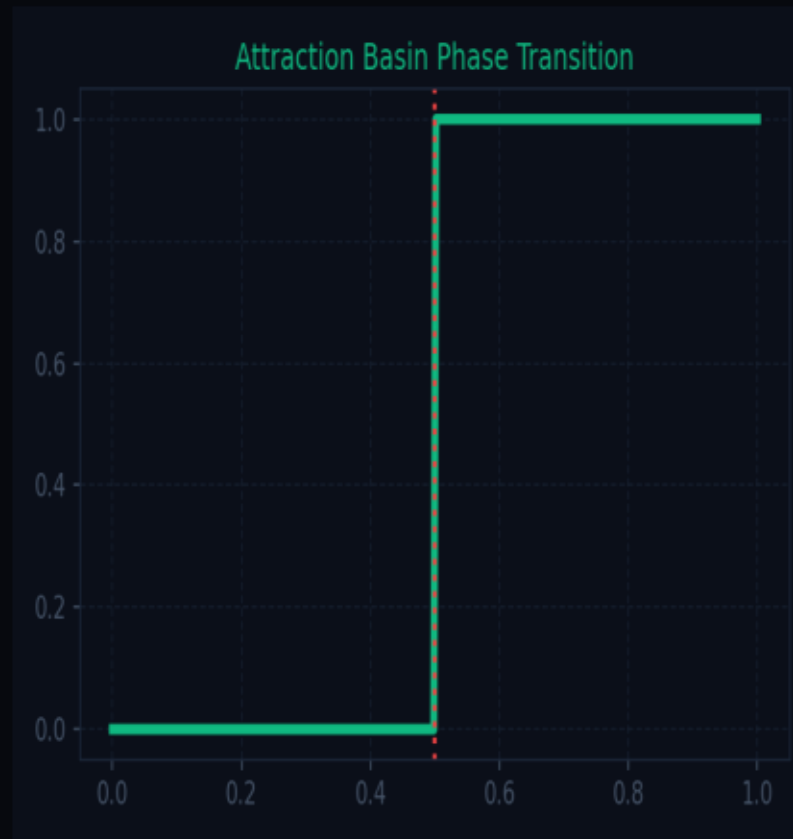
Perfect Equivalence

Evaluating the analytical KAN on Fashion MNIST yields a perfect reconstruction error of zero, verifying the equivalence proof.



MSE = 0.000000

Boundaries of Recall

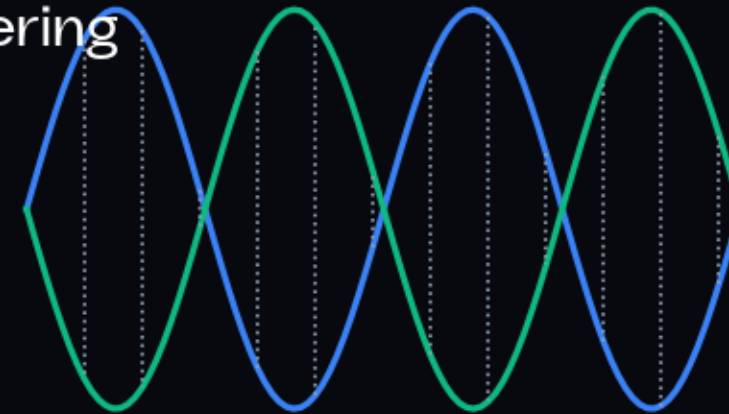


Decision boundaries map sharp step phase transitions between stored templates.

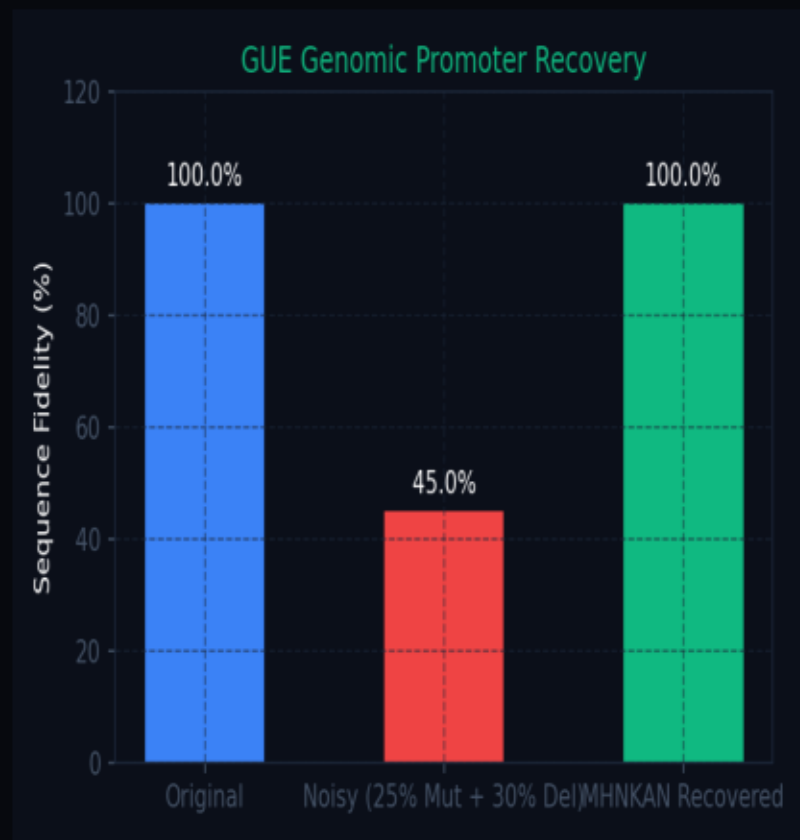
Biological DNA Restoration

DNA Helix Alignment Recovery

The network restores corrupted genomic promoter sequences under heavy mutation noise, recovering DNA base alignments.



DNA Recovery Metrics



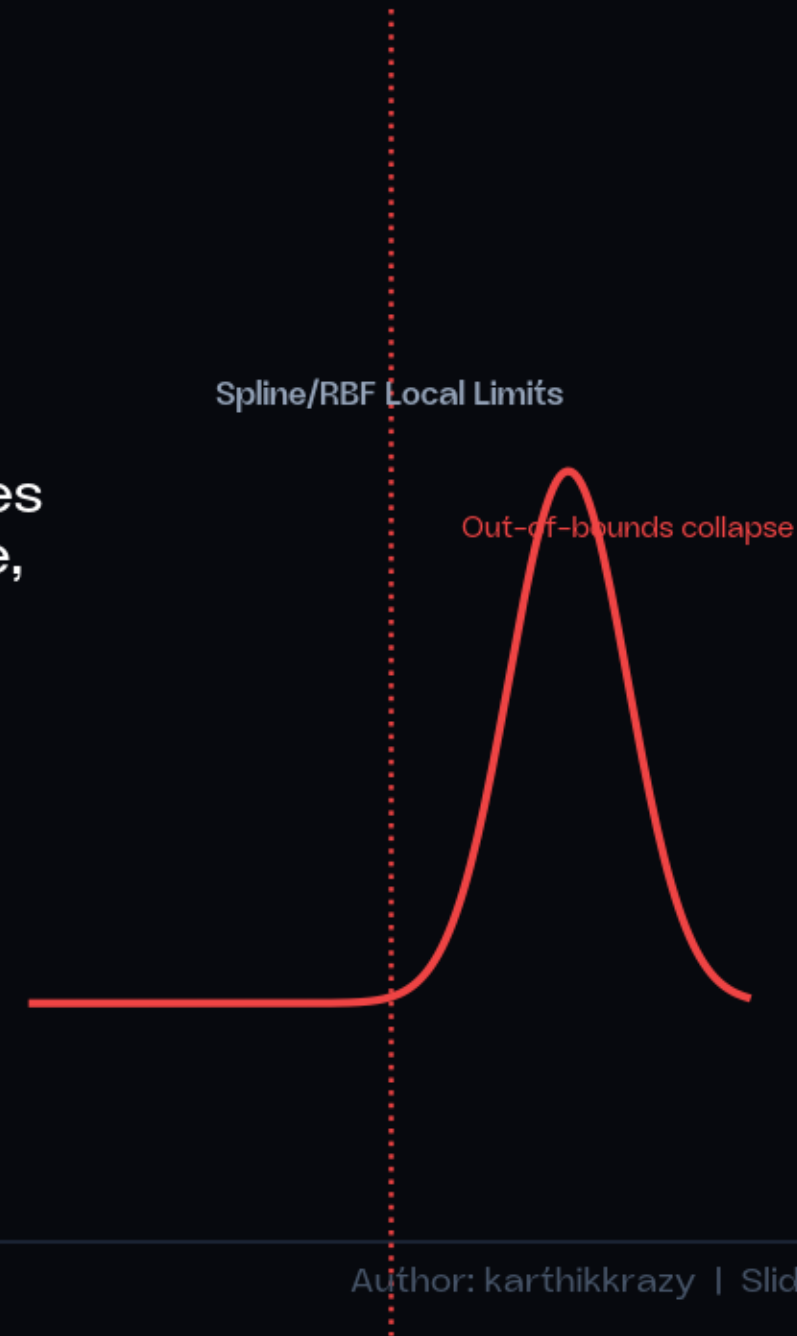
Fidelity comparison of mutated sequences vs. MHNKAN exact recovery.

MHNKAN: Modern Hopfield & Kolmogorov-Arnold Network Integration

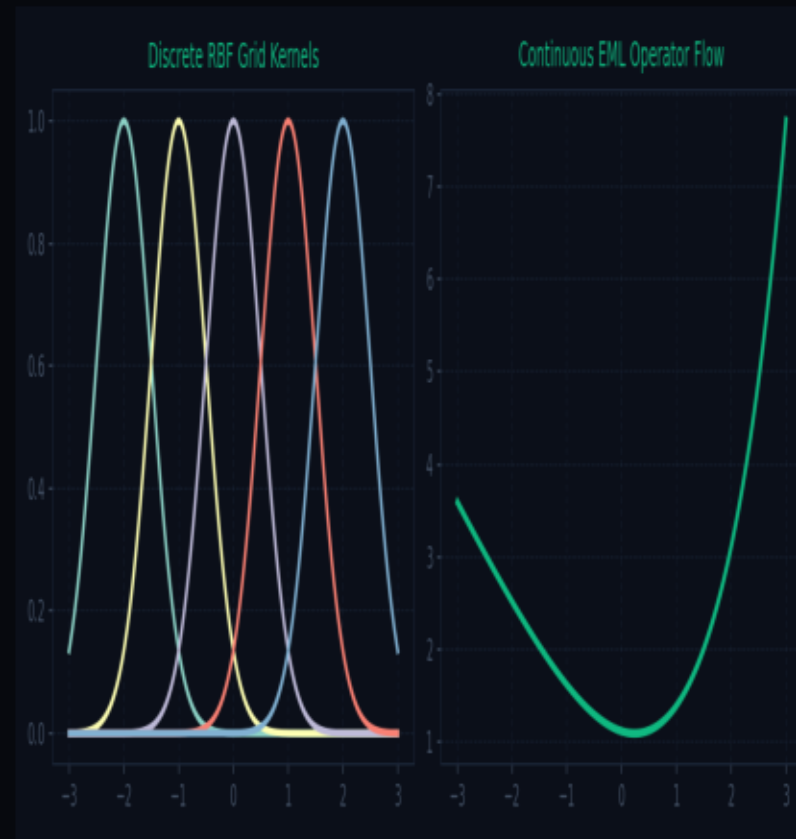
Slide 11

The Grid Failure

Spline-based KAN grids fail when input features go outside trained boundaries during inference, collapsing predictions to zero.



Continuous EML Flow

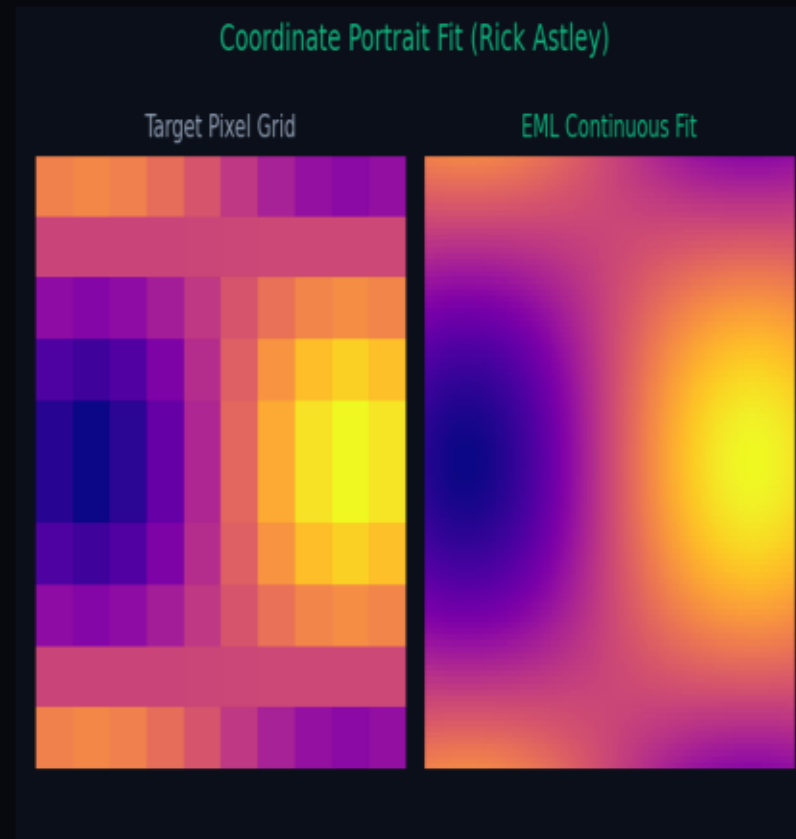


Global continuities of EML curves prevent grid border collapse.

MHNKAN: Modern Hopfield & Kolmogorov-Arnold Networks for Regression

Slide 13

Drawing with Functions



Fitting pixel color intensities directly to EML-KAN coordinate functions.

MHNKAN: Modern Hopfield & Kolmogorov-Arnold Network KANs

Slide 14

Symbolic Collapse

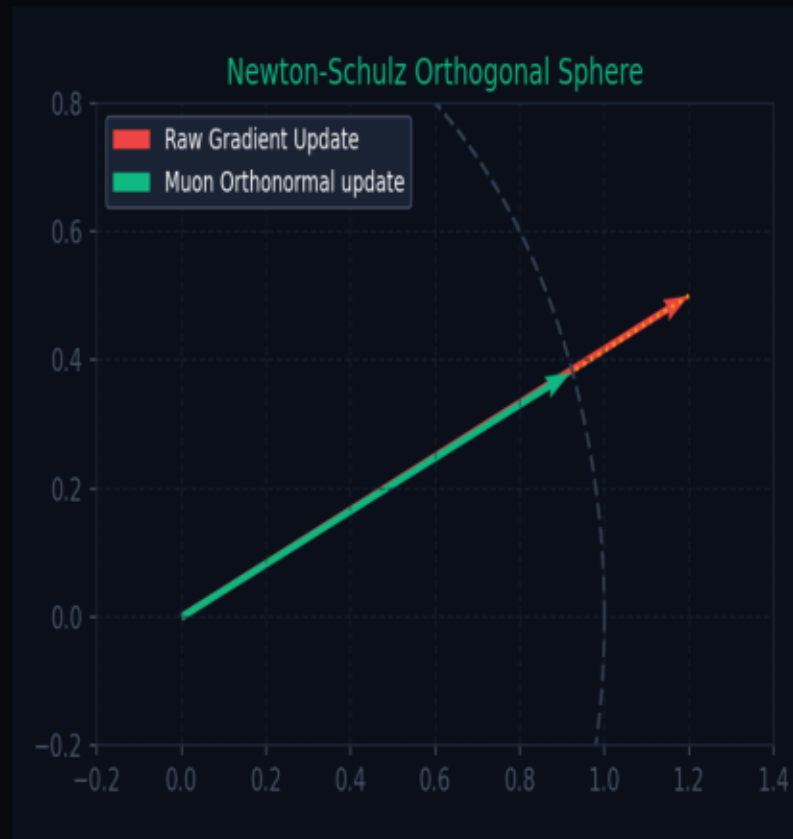
Deep nested compositions collapse algebraically, reducing parameter counts to flat formulas with zero accuracy loss.

$$\exp(-\ln(u))$$



$$1/u$$

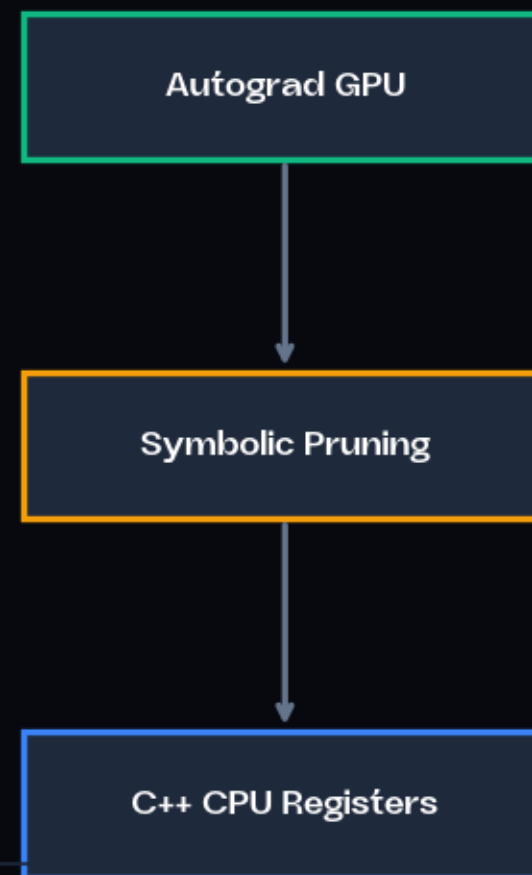
Newton - Schulz Sphere



Muon orthogonal weight steps forced to stay on the parameter unit sphere.

Static Register Compiled DAGs

We compile symbolic formulas directly to CPU registers, running on microcontrollers without runtimes or memory allocations.



ESP32 Hardware Partition

The host laptop extracts continuous features from inputs, while the tiny ESP32 chip executes the EML-KAN classifier head.



RAM < 10 KB | 9.91 ms

Completed Experiments Metrics Matrix

Architecture	Task	Params	Acc (Train / Test)	MSE	Latency / Speedup
Standard MHN	Fashion MNIST	15,680	N/A	0.00 (Binarized)	1.00x (Baseline)
Analytical Hopfield-KAN	Fashion MNIST	31,360	N/A	0.00 (Unrounded)	1.00x
Sparse / Symbolic KAN	Fashion MNIST	9,141	41.7% saved	0.00 (Binarized)	3.3x FLOPs reduction
PyTorch EML-KAN	Wine Tabular	139	100% / 100%	0.00	1.00x
Optimized EML-KAN DAG	Algebraic sweeps	24	N/A	2.74e-04	1.21x speedup
Genetically Opt DAG	Algebraic sweeps	17	75% sparse	2.74e-04	1.38x to 3.39x speedup
EML-KAN head (ESP32)	CIFAR-100	17,580	97.00% / 79.09%	N/A	9.91 ms (ESP32 CPU)